

Hensel's lemma

Given a solution of a polynomial equation in $\mathbb{F}_p$, a question that can be asked is whether this solution can be “lifted” to a solution of the equation in $\mathbb{Z}_p$. Hensel's lemma provides some sufficient conditions under which this can be done. A further generalisation will lift factorisations of polynomials.

Newton-Raphson method

Given a differentiable function f of one variable x , and an “approximate zero” a , one defines y_n iteratively as follows:

$$y_0 = a$$
$$y_{n+1} = y_n - \frac{f(y_n)}{f'(y_n)}$$

where f' denotes the derivative of the function f .

The expectation is that (y_n) is a sequence that (under suitable conditions on f and a) will converge to y such that $f(y) = 0$.

This expectation is driven by the Taylor expansion which says that:

$$0 = f(y) = f(y_n) + (y - y_n)f'(y_n) + e(y, y_n)$$

where $e(y, y_n)$ is expected to go to zero *faster* than $(y - y_n)$ as y_n approaches y . Thus, we see that $f(y_n) + (y - y_n)f'(y_n) \approx 0$. This equation says that y_{n+1} is an approximation to y .

One can give a similar “justification” using the Mean Value Theorem.

One can produce examples of differentiable functions where this method *fails* to produce a convergent sequence. However, if we are working with real numbers, a is close enough to y , the derivatives between a and y are all non-zero of the same sign, then one can prove that the sequence (y_n) converges.

The idea behind one proof of Hensel's lemma is that this works more easily for p -adic numbers than it does for real numbers when we work with polynomial functions.

The p -adic version

Given a polynomial P in $\mathbb{Z}_p[X]$ and a p -adic integer a such that:

- $v_p(P(a)) \geq 1$
- $v_p(P'(a)) = 0$ where P' is the (formal) derivative of P with respect to X .

Note that this means that $P'(a)$ is a unit in $\mathbb{Z}_p$.

We define the sequence y_n inductively as:

$$y_0 = a$$

$$y_{n+1} = y_n - \frac{P(y_n)}{P'(a)}$$

Note that we have *mildly* varied from the above Newton-Raphson iteration by putting $P'(a)$ in the denominator instead of $P'(y_n)$. The latter will also work, but it makes the proof more cumbersome. Since P' is continuous and y_n and a are “close enough”, this change does not matter much.

We claim, by induction on n that $v_p(P(y_n)) \geq (n+1)$ and $v_p(y_n - a) \geq 1$.

Note that we already have $v_p(P(y_0)) = v_p(P(a)) \geq 1$. Thus, we assume, for a given n , that $v_p(P(y_n)) \geq n+1$ and $v_p(y_n - a) \geq 1$ in order to prove that $v_p(P(y_{n+1})) \geq n+2$ and $v_p(y_{n+1} - a) \geq 1$.

First of all we are given that $v_p(P(y_n)) \geq n+1 \geq 1$; so $v_p(y_{n+1} - y_n) \geq 1$. Since $v_p(y_n - a) \geq 1$ by induction, we see that $v_p(y_{n+1} - a) \geq 1$ as well.

We write $y_{n+1} = y_n + h$ where $h = -P(y_n)/P'(a)$. Suppose that $P(X) = \sum_{k=0}^d a_k X^k$. We then have

$$P(y_{n+1}) = P(y_n + h) = \sum_{k=0}^d a_k (y_n + h)^k$$

$$= \sum_{k=0}^d \sum_{r=0}^k a_k y_n^{k-r} h^r \binom{k}{r} = \sum_{r=0}^d \left(\sum_{k=r}^d a_k y_n^{k-r} \binom{k}{r} \right) h^r$$

The term on the right for $r=0$ is $\sum_{k=0}^d a_k y_n^k = P(y_n)$. The term on the right for $r=1$ is $\sum_{k=1}^d a_k y_n^{k-1} k = P'(y_n)$. Shifting these terms to the left we obtain

$$P(y_{n+1}) - P(y_n) - P'(y_n)h = h^2 \sum_{r=2}^d \left(\sum_{k=r}^d a_k y_n^{k-r} \binom{k}{r} \right) h^{r-2}$$

which is h^2 times a p -adic integer which depends on P , y_n and h . In particular, if $v_p(h) \geq n+1$, then we get

$$v_p(P(y_{n+1}) - P(y_n) - P'(y_n)h) \geq 2n+2 \geq n+2$$

Now, $P'(X)$ is also a polynomial in X . So, by a similar argument as above, we note that

$$v_p(P'(y_n) - P'(a)) \geq v_p(y_n - a) \geq 1$$

We now calculate

$$\begin{aligned}
P(y_n) + P'(y_n)h &= P(y_n) - P'(y_n) \frac{P(y_n)}{P'(a)} \\
&= P(y_n) - (P'(a) + (P'(y_n) - P'(a))) \frac{P(y_n)}{P'(a)} \\
&= P(y_n) - P(y_n) + \frac{(P'(y_n) - P'(a)) P(y_n)}{P'(a)} \\
&= \frac{P'(y_n) - P'(a)}{P'(a)} P(y_n)
\end{aligned}$$

Hence, $v_p(P(y_n) + P'(y_n)h) \geq 1 + v_p(P(y_n)) \geq n + 2$.

It follows that $v_p(P(y_{n+1})) \geq n + 2$ as required. Thus, we have proved the claim, by induction.

Since $y_{n+1} - y_n$ is a p -adic integer multiple of $P(y_n)$, we see that $v_p(y_{n+1} - y_n)$ goes to infinity with n . This means that (y_n) is a convergent sequence of p -adic numbers.

Let $\bar{a}$ be the limit of the sequence (y_n) . Since P is a polynomial and thus continuous, we see that $P(y_n)$ converges to $P(\bar{a})$. Since $v_p(P(y_n)) \geq n + 1$, we see that $(P(y_n))$ converges to 0; thus, $P(\bar{a}) = 0$. Since $(y_n - a)$ converges to $\bar{a} - a$ we see that $v_p(\bar{a} - a) \geq 1$.

Given a polynomial $P \in \mathbb{Z}_p[X]$ and $a \in \mathbb{Z}_p$ such that $v_p(P(a)) \geq 1$ and $v_p(P'(a)) = 0$, there is an $\bar{a} \in \mathbb{Z}_p$ such that $P(\bar{a}) = 0$ and $v_p(\bar{a} - a) \geq 1$.

If we think of a as an approximate root (in the p -adic sense) then we have found a p -adic root $\bar{a}$ which is near a (in the p -adic sense).

Factorisation

Given a polynomial P in $\mathbb{Z}_p[X]$ and polynomials Q and R in $\mathbb{Z}_p[X]$ such that:

- $P - QR$ is in $p\mathbb{Z}_p[X]$.
- Q and R do not have a common factor modulo p .

We can ask whether there are polynomials $\tilde{Q}$ and $\tilde{R}$ in $\mathbb{Z}_p[X]$ such that $P = \tilde{Q}\tilde{R}$ in $\mathbb{Z}_p[X]$, where $\tilde{Q} - Q$ and $\tilde{R} - R$ lie in $p\mathbb{Z}_p[X]$.

When $Q = X - a$, then this is very similar to the statement we have just proved.

In view of applications, we restrict our attention to the case when P , Q and R are assumed to be monic. (An argument can be given for the general case, but its intricacies do not concern us here.)

In this case, we can reformulate the question as follows.

Let $\bar{P}, \bar{Q}, \bar{R}$ denote the images of P, Q, R in $\mathbb{F}_p[X]$.

Consider the ring $R = \mathbb{Z}_p[X]/\langle P \rangle$. Note that $R/\langle p \rangle = \mathbb{F}_p[X]/\langle \overline{P} \rangle$. Since $\overline{P} = \overline{Q} \cdot \overline{R}$ as elements of $\mathbb{F}_p[X]$ and, $\gcd(\overline{Q}, \overline{R}) = 1$ in this ring, we can apply the Chinese Remainder Theorem to write:

$$\frac{\mathbb{F}_p[X]}{\langle \overline{P} \rangle} = \frac{\mathbb{F}_p[X]}{\langle \overline{Q} \rangle} \times \frac{\mathbb{F}_p[X]}{\langle \overline{R} \rangle}$$

In fact, suppose $a\overline{Q} + b\overline{R} = 1$ in $\mathbb{F}_p[X]$. To an element (r, s) of the right-hand side one associates the element $t = rb\overline{R} + sa\overline{Q}$ on the left-hand side and checks that $t \bmod \overline{Q} = r$ and $t \bmod \overline{R} = s$.

In particular, if we define $e = b\overline{R}$, then $e \bmod \overline{Q} = 1$ and $e \bmod \overline{R} = 0$. So we see that $e^2 = e$ as an element of $R/\langle p \rangle$.

An element e of a ring R is called an *idempotent*, if it satisfies $e^2 = e$.

Thus, we can look for a “lifting of an idempotent” from $R/\langle p \rangle$ to $R = \mathbb{Z}_p[X]/\langle P \rangle$. (We will later show that this is *sufficient* to get the desired factoring.)

Lifting idempotents

We claim the following result:

Given a ring R and an ideal I and an idempotent e in R/I , there is for each k , an idempotent e_k in R/I^k whose image in R/I is e .

To prove this result, we need only prove it for $k = 2$. For, suppose we have proved it in this case. Then we put $J = I^2$ and apply the result to e_1 and R/J to conclude that there is an idempotent $(e_1)_1$ in R/J^2 as required. Since $J^2 \subset I^3$, we see that the image of $(e_1)_1$ in R/J^2 is the required idempotent there. A simple application of induction then allows us to conclude the result for all k .

On order to simplify notation, we replace R by R/I^2 and pick some lift of e from R/I to R . Thus, we reduce to proving the following:

Given a ring R and an ideal I such that $I^2 = \{0\}$. Given an e in R such that $e^2 - e$ lies in I there is an element n in I such that $\tilde{e} = e + n$ is an idempotent in R .

In other words, we want to find $n \in I$ such that

$$0 = (e + n)^2 - (e + n) = (e^2 - e) + (2en - n)$$

This amounts to solving the equation $(1 - 2e)n = e^2 - e$ to obtain n .

Now we have a bit of magic. Note that

$$(1 - 2e)^2 = 1 - 4e + 4e^2 = 1 + 4(e^2 - e)$$

Since $e^2 - e$ lies in I , we see that $(e^2 - e)I \subseteq I^2 = \{0\}$. It follows that $(1 - 2e)^2 m = m$ for any m in I .

Taking $m = e^2 - e$, we put $n = (1 - 2e)m = (1 - 2e)(e^2 - e)$. The above calculation shows that $(1 - 2e)n = (1 - 2e)^2m = m = e^2 - e$.

In other words, if $\tilde{e} = e + (1 - 2e)(e^2 - e)$, then we get $\tilde{e}^2 - \tilde{e} = 0$ in R . This completes the proof of lifting of idempotents.

When we apply this to $\mathbb{Z}_p[X]/\langle P \rangle$ and the ideal $I = \langle p \rangle$, we obtain idempotents e_k modulo p^k for every n . Taking the p -adic limit of e_k , we obtain an idempotent in $\mathbb{Z}_p[X]/\langle P \rangle$.

Monic polynomials and characteristic polynomials

Given a monic polynomial P in $Z[X]$, for a domain Z , we consider the ring $R = Z[X]/\langle P \rangle$ and let α denote the image of X in R . Let Q denote the field of fractions of Z .

The vector space $V = Q[X]/\langle P \rangle$ (which contains R as a Z -submodule) is spanned by $1, \alpha, \dots, \alpha^{d-1}$. Alternatively, we can consider R as a free module over Z spanned by these elements.

Assume that $P(X) = f_0 + f_1X + \dots + f_{d-1}X^{d-1} + X^d$. Multiplication by α gives an endomorphism M_α of this vector space V which is represented by the companion matrix:

$$C_P = \begin{pmatrix} 0 & 0 & \dots & 0 & -f_0 \\ 1 & 0 & \dots & 0 & -f_1 \\ 0 & 1 & \dots & 0 & -f_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -f_{d-1} \end{pmatrix}$$

Exercise: Check that P is the characteristic polynomial of this matrix.

In other words, P is the characteristic polynomial of the endomorphism M_α of V . Note that this statement is independent of the choice of basis.

Block form and idempotents. Multiplication by an idempotent e in R gives an idempotent endomorphism E of the vector space V .

Since $E^2 = E$, we see that E can be diagonalised. More explicitly, let $\{u_1, \dots, u_r\}$ be a basis of the image U of E and $\{w_1, \dots, w_s\}$ be a basis of the kernel W of E . Given any vector $v \in V$, we can write $v = Ev + (v - Ev)$ and note that $E(v - Ev) = Ev - E^2v = 0$. Moreover, if $E(E(v)) = 0$, the $E(v)$ is 0. It follows that $\{u_1, \dots, u_r, w_1, \dots, w_s\}$ is a basis of V .

Since R is commutative we see that E and M_α commute, so that $M_\alpha E(v) = EM_\alpha(v)$. Thus, M_α takes U to U and W to W . Thus, in the above basis of V , we have the block form for M_α

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

It follows that the characteristic polynomial of M_α can be written as the product of the characteristic polynomial of A and the characteristic polynomial of B .

This, gives a factoring of P in $K[X]$.

Integrality. In the specific case which we are working with, e is an element of R , hence $e(R)$ is contained in R . Since $\mathbb{Z}_p$ is a principal ideal domain, one can choose $u_1, \dots, u_r$ which give a basis of $e(R)$ as a module over $\mathbb{Z}_p$. Similarly, $w_1, \dots, w_s$ can be chosen to be a basis of $(1 - e)R$ over $\mathbb{Z}_p$. It follows that A and B have entries in $\mathbb{Z}_p$. Thus, we can conclude the following.

Given a monic polynomial P in $\mathbb{Z}_p[X]$ and an idempotent element e in $\mathbb{Z}_p[X]/\langle P \rangle$, we can write $P = QR$ so that there is a decomposition:

$$\frac{\mathbb{Z}_p[X]}{\langle P \rangle} = \frac{\mathbb{Z}_p[X]}{\langle Q \rangle} \times \frac{\mathbb{Z}_p[X]}{\langle R \rangle}$$

where e corresponds to the idempotent $(1, 0)$ on the right-hand side.

When the idempotent e has been obtained by the above procedure to lift the idempotent associated with a factoring of P modulo p , we see that this factoring of P is a lift of the given factoring of P modulo p .

Multiple factors

Given a polynomial $P \in \mathbb{Z}_p[X]$ consider its image $\bar{P}$ in $\mathbb{F}_p[X]$. Since the latter is a UFD, we can write

$$\bar{P} = \bar{Q}_1^{r_1} \cdots \bar{Q}_k^{r_k}$$

where $\bar{Q}_i$ are *distinct* irreducible polynomials in $\mathbb{F}_p[X]$ and r_i are positive integers.

By the Chinese Remainder Theorem, We have idempotents $\bar{e}_1, \dots, \bar{e}_k$ in the ring $\mathbb{F}_p[X]/\langle \bar{P} \rangle$ which correspond to a decomposition:

$$\frac{\mathbb{F}_p[X]}{\langle \bar{P} \rangle} = \frac{\mathbb{F}_p[X]}{\langle \bar{Q}_1^{r_1} \rangle} \times \cdots \times \frac{\mathbb{F}_p[X]}{\langle \bar{Q}_k^{r_k} \rangle}$$

By Hensel's lemma, these idempotents lift to idempotents $e_1, \dots, e_k$ in the ring $\mathbb{Z}_p[X]/\langle P \rangle$. The above discussion then shows that this leads to a decomposition:

$$\frac{\mathbb{Z}_p[X]}{\langle P \rangle} = \frac{\mathbb{Z}_p[X]}{\langle P_1 \rangle} \times \cdots \times \frac{\mathbb{Z}_p[X]}{\langle P_k \rangle}$$

where $P = P_1 \cdots P_k$ in $\mathbb{Z}_p[X]$ and the image of P_i in $\mathbb{F}_p[X]$ is $\bar{Q}_i^{r_i}$. Furthermore, this leads to a decomposition:

$$\frac{\mathbb{Q}_p[X]}{\langle P \rangle} = \frac{\mathbb{Q}_p[X]}{\langle P_1 \rangle} \times \cdots \times \frac{\mathbb{Q}_p[X]}{\langle P_k \rangle}$$

Note: The polynomials P_i *need not* be irreducible. However, we do know that P_i and P_j have no common factor if $i \neq j$.

Unramified case

Given that P is a monic polynomial of degree d with coefficients in $\mathbb{Z}_p$ such that $\bar{P} = P \bmod p$ is irreducible in $\mathbb{F}_p[X]$.

It follows that P is irreducible in $\mathbb{Z}_p[X]$ as a factoring of P would yield a factoring of $\bar{P}$.

Since $\mathbb{Z}_p$ is a UFD this means that P is irreducible in $\mathbb{Q}_p[X]$. Thus, $K = \mathbb{Q}_p[X]/\langle P \rangle$ is a field which is spanned by $1, \alpha, \dots, \alpha^{d-1}$ as a vector space over $\mathbb{Q}_p$; here α denotes the image of X in K . In fact, we have seen that $R = \mathbb{Z}_p[X]/\langle f \rangle$ can be identified with the $\mathbb{Z}_p$ span of these elements of K .

Note that $R/\langle p \rangle$ can be identified with $\mathbb{F}_p[X]/\langle \bar{P} \rangle$ which is a field. This shows that p is a prime in R .

If P is a monic polynomial in $\mathbb{Z}_p[X]$ whose image in $\mathbb{F}_p[X]$ is irreducible, then $R = \mathbb{Z}_p[X]/\langle P \rangle$ is a domain in which p is a prime.

Now consider a monic polynomial $P \in \mathbb{Z}_p[X]$ which splits into *distinct* irreducible factors modulo p . In other words, $\bar{P} = \bar{P}_1 \cdots \bar{P}_k$ where $\bar{P}_i$ are irreducible in $\mathbb{F}_p[X]$.

As seen above this means that $P = P_1 \cdots P_k$ in $\mathbb{Z}_p[X]$ where the image of P_i in $\mathbb{F}_p[X]$ is $\bar{P}_i$ which is irreducible. It follows that P_i is irreducible in $\mathbb{Z}_p[X]$ and these factors are mutually co-prime (since $\bar{P}_i$ are distinct). Further, We have a decomposition:

$$R = \frac{\mathbb{Z}_p[X]}{\langle P \rangle} = \frac{\mathbb{Z}_p[X]}{\langle P_1 \rangle} \times \cdots \times \frac{\mathbb{Z}_p[X]}{\langle P_k \rangle} = R_1 \times \cdots \times R_k$$

where $R_i = \mathbb{Z}_p[X]/\langle P_i \rangle$ for $i = 1, \dots, k$. For each $\alpha \in R$, let $(\alpha_1, \dots, \alpha_k)$ be its image in the right-hand side.

For each i , we use π_i to denote the element of R whose image in the right-hand side is p in R_i and 1 in R_j for $j \neq i$. (In terms of the idempotents e_i , we have $\pi_i = pe_i + \sum_{j \neq i} e_j$.) Then, $R/\langle \pi_i \rangle$ is naturally isomorphic to $R_i/\langle p \rangle$ which is a field. Thus, π_i is a prime element of R (here we are extending the notion of primes to R which is not a domain). Note that we have the identity $p = \pi_1 \cdots \pi_k$ in R .

Recall that the ring $\mathbb{Z}_{(p)}$ consists of elements a/b in $\mathbb{Q}$ where b is not divisible by p . We now further assume that P has coefficients in $\mathbb{Z}_{(p)}$. In this case, we have a subring $T = \mathbb{Z}_{(p)}[X]/\langle f \rangle$ of R which is spanned by $1, \alpha, \dots, \alpha^{d-1}$ over $\mathbb{Z}_{(p)}$. Since $\mathbb{Z}_{(p)}$ is dense in $\mathbb{Z}_p$ in the p -adic metric, we see that T is dense in R .

The element π_i lies in the open set U_i of R which consists of α such that α_i lies in $p + p^2 R_i$ and α_j lies in $1 + p R_j$ for $j \neq i$. Since T is dense in R , we see that there is an element q_i in $U_i \cap S_0$. It follows that $R/\langle q_i \rangle = R_i/\langle p \rangle$. In fact, these are finite, so we see that $T/\langle q_i \rangle = S/\langle q_i \rangle$ as a dense subset of a finite set is the whole set! This shows that q_i is a prime in T . Moreover, $q_i = u_i \pi_i$ for some unit

u_i in R . It follows that $q_1 \cdots q_k = u \cdot p$ for some unit u in R . Since $q_1, \dots, q_k$ and p are in T , it follows that u is a unit in T .

Given a monic polynomial P in $\mathbb{Z}_{(p)}[X]$ which splits into k distinct irreducible factors modulo p we get a factoring $p = v \cdot q_1 \cdots q_k$ of p in $T = \mathbb{Z}_{(p)}[X]/\langle P \rangle$ where q_i are primes in T and v is a unit in T .

Note: When P is in $\mathbb{Z}[X]$, the ring $T_0 = \mathbb{Z}[X]/\langle P \rangle$ is dense in R , so we can even take $q_i \in T_0$. *However*, these are only prime in T and v is only a unit in T .

Note: Even though f factorises in $\mathbb{Z}_p[X]$ it *may* be irreducible in $\mathbb{Z}_{(p)}[X]$ since Hensel's lemma requires us to take limits. Thus, even though R is *not* a domain, T and T_0 *could* be domains. In fact, this will be the case of interest to us as we go along.

General extensions

If P is an irreducible monic polynomial in $\mathbb{Z}_p[X]$, then as a consequence of the above results, its image $\overline{P}$ in $\mathbb{F}_p[X]$ is of the form $\overline{Q}^r$ for an irreducible monic polynomial $\overline{Q}$ of degree s in $\mathbb{F}_p[X]$ where $d = rs$.

Since $\mathbb{Z}_p$ is a UFD, the polynomial P is also prime in $\mathbb{Z}_p[X]$ and $\mathbb{Q}_p[X]$. Thus, $R = \mathbb{Z}_p[X]/\langle P \rangle$ is a domain and $K = \mathbb{Q}_p[X]/\langle P \rangle$ is a field.

Now, $R/\langle p \rangle = \mathbb{F}_p[X]/\langle \overline{Q}^e \rangle$ has a unique prime (in fact, maximal) ideal

$$m = \frac{\langle \overline{Q} \rangle}{\langle \overline{Q}^r \rangle}$$

Thus, R has a unique maximal ideal M which contains p ; the image of M in $R/\langle p \rangle$ is m . We see that $M = \langle p, \beta \rangle$ where β is some element of R whose image in $R/\langle p \rangle$ is $\overline{Q}$.

Now $\overline{Q}^r$ becomes 0 in $R/\langle p \rangle$, which shows that $M^r \subset pR$. The sequence of ideals in R

$$M \supset M^2 \supset \cdots \supset M^r$$

has the images

$$m \supset m^2 \supset \cdots \supset \{0\}$$

in $R/\langle p \rangle$. Since these images are distinct, we obtain isomorphisms $M^i/M^{i+1} \cong m^i/m^{i+1}$ for $i = 1, \dots, (r-1)$.

In particular, we see that M/M^2 is a 1-dimensional vector space over $R/M \cong \mathbb{F}_p[X]/\langle \overline{Q} \rangle$. It follows easily that if $\pi \in M \setminus M^2$ is any element, then $M = \pi R$ and $pR = M^r = \pi^r R$.

Recall that $R = \mathbb{Z}_p \oplus \mathbb{Z}_p \alpha \oplus \cdots \oplus \mathbb{Z}_p \alpha^{d-1}$ where α is the image of X in R and d is the degree of P . Given any non-zero element γ in R , it satisfies the characteristic polynomial of the matrix $M \in M_d(\mathbb{Z}_p)$ that gives the multiplication by γ in this basis of R . Since $K = \mathbb{Q}_p[X]/\langle P \rangle$ is a field, multiplication by γ is invertible.

Thus, M has a non-zero determinant $\det(M)$. We note that $\det(M - \gamma I_d) = 0$ shows that $\det(M)$ lies in $\gamma R \cap \mathbb{Z}_p$.

This shows that any non-zero ideal I in R has a non-zero intersection with $\mathbb{Z}_p$. Thus, we see that $I \cap \mathbb{Z}_p = p^k \mathbb{Z}_p$ for some positive integer k . It follows that I contains M^{ke} . It follows that any ideal I in R is of the form $\pi^a R$ for some positive integer b .

We can summarize the above as follows:

Given an irreducible monic polynomial P of degree d in $\mathbb{Z}_p[X]$, the domain $R = \mathbb{Z}_p[X]/\langle P \rangle$ has a unique maximal ideal of the form πR such that $R/\langle \pi \rangle$ is isomorphic to the finite field $\mathbb{F}_{p^f}$; we have $d = rs$ and $p = u \cdot \pi^r$ where u is a unit in R . Moreover, every ideal in R is of the form $\pi^a R$.